AeroSentinel

TEAM ASTRO / Design document / Final validated runtime

Purpose and environment contract

AeroSentinel evaluates a tool-using agent that diagnoses and repairs a simplified CubeSat mission. The environment supplies seeded faults, files, live telemetry and protected success targets. A separate verifier scores the resulting mission history. This project implements an evaluation environment and does not train a policy.

Persistent state includes elapsed time, battery energy, temperature, scalar pointing error, delivered data, payload energy and useful exposure. The tools list/read/write files, run Python, advance orbit and finish. An ordinary successful tool call consumes 30 simulated seconds. Orbit steps advance 1–30 minutes within a 5,700-second mission. The harness integrates at sunlight, contact and observation boundaries.

Task progression and nontrivial decisions

Task group	Reasoning required
Five guided repairs: EPS, ADCS, thermal, RF and cascade	Inspect the manual, identify seeded faults and restore safe operating behavior.
RF energy tradeoff (hard)	Choose throughput and transmitter power that satisfy data, link-margin and energy constraints.
Early pass recovery (hard)	Repair pointing and radio settings before an irreversible contact deadline.
Observation scheduling (hard)	Change payload duty over time to collect exposure only in the useful window while respecting total energy.

A static observation duty must be at least 0.90 to satisfy exposure. Whole-mission energy permits at most 0.126–0.189 for the generated window lengths. These constraints cannot both hold. A feasible schedule disables the instrument before the window, enables it during exposure, then disables it. Tests accept alternative feasible controllers and schedules.

Declared simulation assumptions

Solar power depends on calibration, fixed array efficiency/area, irradiance and cosine pointing. The battery integrates generation minus load in Wh. Bus voltage is linear in state of charge. Pointing follows a scalar drift/damping model. Temperature uses lumped heat capacity and linear heat loss. Radio margin includes bandwidth and pointing penalties; data accumulates only during contact. Payload energy accrues outside the exposure window as well as inside it.

This model has no orbital propagator, cell chemistry, three-axis rigid-body dynamics or flight validation. Legacy charger-rate, actuator-frequency and polling fields are descriptive and do not affect score. Exact equations and controls are documented in docs/model-assumptions.md.

Verification and evidence

Protected outcomes, bounded execution and honest reporting

Reward and termination

Standard raw weights are power 0.35, thermal 0.25, pointing 0.20 and communications 0.20. Observation tasks scale these weights by 0.8 and add useful exposure 0.20. Progress equals clamp((raw – initial no-op raw)/(1 – initial no-op raw), 0, 1). Every required mission check must pass independently. A failed check prevents a success score of 0.95 or above; catastrophic events remain latched and score zero. Final scoring validates the current files and projects remaining mission time without rewriting accumulated history.

Isolation, limits and portability

The harness owns verifier code, reward and protected mission facts. Agent workers run in the sandbox as UID 1001; behavioral fixtures run as UID 1002. Authenticated controls, read-only root, an internal network, bounded workspace/output/CPU/memory, a 128-process cgroup and descendant cleanup limit access. Docker and the host kernel remain trusted. Finite tests do not prove absence of every escape or reward exploit.

The runner allows 40 received replies, 10 tools per reply and 600 seconds. Tool operations have a 60-second budget; final verification has a separate 120-second budget. Baselines share 120 seconds per task and must pass before an API call. Provider calls run in cancellable processes. At most two logged transient retries share the episode budget. Received malformed replies consume a step; a failed request with no reply does not. Model, key, endpoint and exact UTF-8 append text are configurable with a common interface.

Evidence on the final runtime fingerprint

Check	Observed result
Linux regressions and reset sweep	71 tests passed; 160 task/seed combinations across eight task families.
Live isolation / all-task gate	12 container tests passed; all 8 reference = 1.0 and no-op = 0.0.
Scripted policy comparison, 40 episodes per policy	No-op 0; nominal 25; maximum radio 20; privileged reference 40 successes. Every non-reference shortcut failed the 15 hard-task episodes.
Four scripted attack probes	Relaxed pointing 0.800; forged telemetry 0.000; missed contact 0.418; missed observation 0.800. All terminate as failure.
Four final-source organizer-provider attempts	EPS succeeds with and without append. Early-pass and observation attempts terminate on APIConnectionError after bounded retries. No hard-task model success claim.

Run A: EPS seed 0, no append, reward 1.0, 7 replies, 29.31 harness seconds. Three attempts received replies and qualify under the strict complete-run packager; the zero-reply observation attempt remains diagnostic evidence. Elapsed values use a monotonic harness clock. Host/VM pauses caused large UTC gaps, so these are not independent external wall-clock measurements. Raw inputs, replies, actions, errors, state hashes, rewards and final reasons remain in source-matched logs. Historical results do not certify this version.

Reproduction and remaining submission requirements

One Compose command builds and runs the selected task. CI checks the isolated empty-cache 180-second build budget and offline/live regressions. Packaging verifies source fingerprints, complete episodes, PDF presence, checksums and credential scans. The required archive is Astro.zip. Public repository visibility is an explicit user choice and conflicts with the PS private-repository requirement. Exact Astro/astro case, organizer access, deadline and form upload remain unconfirmed.

Evidence: submission_bundle/validation-evidence.json. Harness SHA-256: 9e2d94e2c2473ccc962847d2… Sandbox: 2e0b2933cd17ebf44fe04559…